NOTE A data warehouse, lake, lakehouse, or fabric can be complementary solutions. These architectures are not necessarily alternatives, but they could be treated as complementary solutions. You simply need to navigate away from their standard socio-technical architectures and align them with data mesh's socio-technical architecture. The way to do that is to decentralize ownership.

Each architecture we describe here comes in many flavors. In each organization, they undergo many tweaks, adjustments, and improvements. But we will keep our description simple and will not consider all possible variations. In our comparison, we will refer to the standard, most popular ways of their implementation.

2.2.1 Enterprise data warehouse

The *enterprise data warehouse* is an approach to storing data in the central data repository that the central data team owns. Data is ingested to the data warehouse by using the extract, transform, load (ETL) or extract, load, transform (ELT) approach, usually directly from systems-of-record databases. Data is harmonized into a canonical model that is stored in the data warehouse. Data is either directly consumed from the data warehouse or through simpler data marts usually focused on just one business function (figure 2.3).

